Layered Typography-Aware Poster Generation with Editable Text

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Abstract

Text-to-image diffusion models can synthesize visually rich poster-like images, but they remain unreliable when the image must contain exact, editable, and well-composed text. We study this failure mode in text-heavy poster generation and propose a layered pipeline that separates visual synthesis from typography. Qwen-Image first generates a no-text background; a saliency- and contrast-aware layout module places structured text boxes; a learned typography-tool router selects fonts and display effects; and a deterministic renderer produces the final typography layer with strokes, shadows, gradients, local backing, and editable layout metadata. Because the background is fixed once generated, edits to typography never trigger regeneration — and therefore never alter the underlying scene. On 24 poster-style prompts, our renderer achieves mean outside-text SSIM 0.997 and LPIPS 0.0049 across edited variants, while strong T2I baselines that re-prompt for new text routinely change the depicted subject (composition, identity, lighting). Our results suggest that treating typography as a structured layer is a practical path toward controllable text-heavy visual synthesis.

1. Introduction

Modern text-to-image (T2I) diffusion models such as latent diffusion models [9], Imagen [10], and DALL·E-style systems [1] can generate compelling images from natural-language prompts. Yet posters, advertisements, book covers, flyers, and social media graphics expose a persistent weakness: image generators often produce plausible visual compositions but fail to render precise text. Letters become pseudo-glyphs, words are misspelled, captions appear in the wrong place, and typography cannot be edited after generation. This is not only an aesthetic problem. Text-heavy visual synthesis requires exact symbolic content, spatial grounding, visual hierarchy, and downstream editability.

We address this setting through *layered typography-aware poster generation*. Instead of asking a diffusion model to jointly invent imagery and rasterize text, we split the problem into background synthesis and structured typography. The model generates the visual scene without text; a layout planner selects candidate text regions; and a renderer writes exact strings as an editable text layer. The final output is both a raster poster and a JSON layout containing text, bounding boxes, colors, and roles. The key practical consequence is that the visual scene becomes a stable artifact: once a user accepts a background, subsequent text edits are cheap, deterministic, and structurally guaranteed to leave the scene unchanged. By contrast, re-prompting a T2I model with new text regenerates the entire image, so the scene drifts — sometimes drastically, as when the depicted subject changes identity or composition between edits. This design is deliberately pragmatic: it uses diffusion models for the component where they are strongest and uses structured rendering for the component where exactness and stability matter most.

The project makes three contributions. First, we implement an end-to-end pipeline for poster generation with editable text layers. Second, we train compact modules: a layout ranker, a GenPoster-trained text-box prior, and a typography-tool router that chooses among multiple font/effect presets. Third, we evaluate with edit-preservation metrics, layout validity, style-router accuracy against a majority baseline, style diversity figures, and qualitative comparisons to direct T2I prompting. In the final version, we replace the early black-band typography with open-source display fonts and reusable effects such as neon glow, foil gradients, retro shadows, block shadows, letterpress, and stamp-like rendering. Although the approach is not an end-to-end text-rendering diffusion model, it produces a substantially more reliable system for the specific design task we target.

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2. Related Work

Text rendering in image generation. Methods such as TextDiffuser [2], AnyText [12], and Glyph-ByT5 [8] improve visual text rendering by injecting glyph- or character-level information into diffusion models. More recent text-capable backbones such as Qwen-Image render legible typography directly. These works show that explicit text supervision is important and that modern T2I models are no longer broken at text legibility. Two limitations remain. First, their outputs are raster images whose typography is not represented as a persistent editable layer: changing a single word requires re-running the generator. Second, regeneration does not preserve the surrounding scene — re-prompting drifts composition, lighting, and even the identity of depicted subjects (§5, Fig. 2). Our system makes typography a first-class structured object and treats the background as a fixed asset, addressing both limitations simultaneously.

Layout-aware generation. GLIGEN [6], ControlNet [14], and LayoutGPT [4] condition generation on spatial layouts or object boxes. These approaches provide more control than prompt-only generation, but the controlled elements are usually objects or semantic regions rather than typographic hierarchy. Our layout variables are text-specific: role, bounding box, color, and renderable string content.

Graphic and poster layout. PosterLayout [5], CGL-GAN [15], and AutoPoster [7] study visual-textual presentation design and content-aware layout. Our work is closer to this line than to pure scene synthesis: the goal is a usable design artifact. However, we combine generated backgrounds, structured typographic rendering, and edit-preservation evaluation in a compact pipeline.

3. Method

Given a prompt p and text strings $\mathcal{T} = \{t_i\}_{i=1}^{n}$, our goal is to produce a poster image I and a structured layout $L = \{(t_i, b_i, r_i, c_i)\}_{i=1}^{n}$, where b_i is a bounding box, r_i is a role such as title or metadata, and c_i is a foreground color.

Background generation. For each poster prompt, we generate a background image using Qwen-Image. The background prompt describes only the visual scene and explicitly excludes text, letters, watermarks, signage, numbers, and logos. For comparison, we also generate a direct T2I baseline with the same base model, but its prompt asks the model to render the exact requested text inside the poster. Thus the comparison isolates two uses of the same image generator: direct rasterized poster generation versus no-text background synthesis followed by structured typography.

Candidate layout scoring. For title, subtitle, and metadata roles, we enumerate candidate boxes at several vertical anchors and horizontal offsets. Each candidate receives a hand-designed score:

$$s(b) = -E(b) - \lambda_o O(b) - \lambda_y |y_b - y_r| + \lambda_m M(b), \quad (1)$$

where $E(b)$ is mean Sobel edge density inside the box, $O(b)$ penalizes overlap with previously selected text boxes, y_b is the box center, y_r is the role-specific preferred vertical location, and $M(b)$ is a margin bonus. Low edge density is a lightweight proxy for saliency and background clutter: placing text on low-edge regions tends to improve readability and avoids covering detailed visual content.

Learned layout ranker. We train a small MLP to predict the normalized candidate score. Each candidate is represented by 14 features: normalized (x, y, w, h), normalized area, local edge mean and standard deviation, local luminance mean and standard deviation, normalized margin, distance to role-preferred vertical position, and a three-dimensional role one-hot vector. The network has two hidden layers with 64 and 32 units and is trained with mean-squared error. This ranker is not required for the deterministic fallback, but it provides a learned layout component and a natural bridge toward larger learned design models.

Typography-tool router. The first version of the system used a single high-contrast black-band style, which improved OCR but made outputs visually repetitive. We therefore add a tool registry with 19 typography presets: classic travel, cinematic, premium advertisement, Bauhaus, neon, literary, technology, sport, editorial serif, modern minimal, playful social, clean corporate, organic market, luxury fashion, retro pop, handmade marker, script invite, brutalist type, and space age. Each preset selects a layout template, font family, color palette, stroke width, shadow, tracking, backing mode, and display effect. The effects are implemented as deterministic rendering tools, not opaque online APIs: neon glow, foil or soft gradients, retro offset shadows, block shadows, letterpress, and stamp-like overlays. A small router predicts the preset from prompt keywords, category, text-length statistics, and weak design labels.

Spatial refinement. The final system adds a template-aware spatial refinement pass. Candidate boxes are sized as a function of text length, clamped to safe margins, and augmented with poster-like anchor layouts. The scorer uses template-specific preferred vertical positions instead of one global title preference, and it penalizes both overlap and

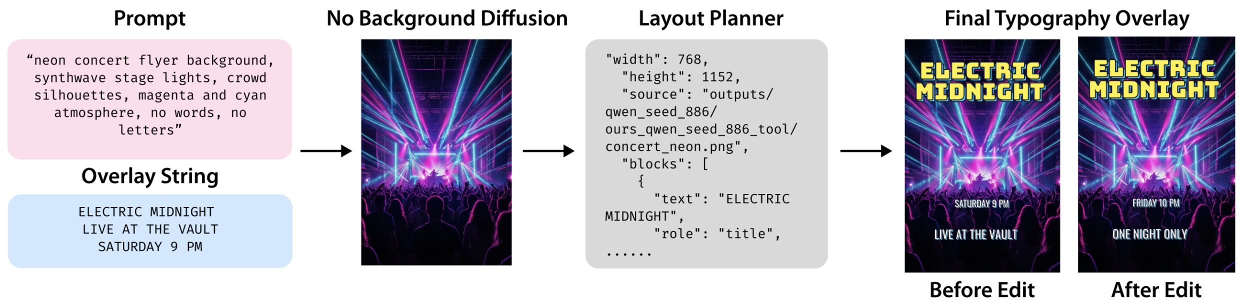

Figure 1. **Pipeline.** We generate a no-text visual background, plan structured typography using saliency and contrast cues, render exact text deterministically, and preserve the layout as editable JSON metadata.

near-collisions between rendered text regions. This prevents bottom-title, editorial, and left-column styles from collapsing to the same layout and improves font-size hierarchy.

Poster-data layout prior. To move beyond hand-authored templates, we train on the full 102K-record GenPoster-100K annotation release [3]. Each record contains text-layer metadata including string content, bounding boxes, canvas size, font family, font size, and alignment. We construct positive examples from real text boxes and five random negatives per positive, then train a layout MLP with binary cross entropy. We also filter pathological skinny boxes and cluster per-style role boxes into reusable normalized priors. At rendering time, the planner combines these learned priors with saliency and contrast scoring, while enforcing a minimum title footprint so the learned prior does not collapse titles into small corner labels.

Typography rendering and editing. The selected text boxes are rendered with a deterministic FreeType-based renderer. The renderer fits font size to the box, wraps long strings, adds a high-contrast stroke, shadow, and translucent local backing band, and saves both the image and layout JSON. Editing is performed by changing the text strings and re-rendering only the typography layer on the same background.

4. Experiments

Dataset. We use 24 poster-style prompts spanning travel, film, product advertising, museum exhibitions, concerts, books, restaurants, technology events, fitness, education, beauty, music, food markets, architecture, transportation, games, fashion, photography, charity, board games, and cookbooks. Each prompt contains three text strings and an edited variant. For the final style/layout upgrade, we additionally train from GenPoster-100K metadata: 102,703 poster annotations, 102,699 with text layers, and 3,604,506 positive/negative layout examples.

Table 1. Main quantitative diagnostics over 24 prompts and edited variants. Edit preservation is measured outside the union of before/after text-box masks.

Metric	Value
Layout validity[1]	1.000
Contrast proxy	0.526
Edit outside-mask SSIM $\uparrow$	0.997
Edit outside-mask LPIPS $\downarrow$	0.0049
Edited prompt pairs	24

Baselines. The main baseline is direct T2I generation with the same Qwen-Image model. The baseline prompt asks the model to produce the poster with the exact requested text. This is a strong practical baseline because it matches how users commonly attempt text-heavy image generation: one prompt requests both imagery and typography.

Metrics. We report rendered text exactness, layout validity, a contrast proxy, style-router accuracy, and edit background preservation. Layout validity is the fraction of boxes that are in-bounds and non-overlapping. Edit preservation is measured across the 24 prompt/edit pairs by masking the union of before- and after-edit text boxes, then computing SSIM [13] and LPIPS outside the masked regions. We treat OCR as a diagnostic rather than a primary metric because Tesseract [11] is sensitive to local installation, stylized display fonts, and textured poster backgrounds.

[1]Layout validity is enforced by the planner; the reported value indicates that the planner produced a valid in-bounds non-overlapping configuration on every prompt in the test set, i.e. the search never failed to find a feasible layout.

Table 2. Learned layout and typography-tool priors. The router uses weak pseudo-labels, so its accuracy measures tool-selection consistency rather than human aesthetic preference.

Quantity	Value
Poster annotations for layout	102,703
Layout examples	3,604,506
Layout validation accuracy	0.975
Router training examples	99,996
Typography-tool presets	19
Router validation accuracy	0.789
Majority-class baseline	0.398

Table 3. Layout ranker training summary. The validation correlation indicates that the lightweight MLP learns the candidate-box scoring objective.

Quantity	Value
Candidate boxes	1248
Train / validation	998 / 250
Feature dimension	14
Epochs	450
Final validation MSE	0.200
Final validation correlation	0.887

5. Results

Our main empirical claim is edit preservation. Table 1 reports mean SSIM and LPIPS outside the union of before- and after-edit text-box masks across 24 prompt/edit pairs. The outside-mask SSIM is 0.997 and LPIPS is 0.0049, showing that changing a title, subtitle, date, or callout leaves the generated background effectively unchanged. Because the final text is rendered from structured strings, rendered text exactness is 1.000 by construction; we therefore treat exactness as a system property rather than the central quantitative result.

The practical importance of this number is that re-prompting a strong T2I model with new text does not, in general, preserve the depicted scene. In our 24-prompt comparison, the direct Qwen baseline frequently regenerates non-typographic content when only the strings change: subjects shift pose, lighting changes, and in several cases the depicted entity itself is replaced. In the Solar Wake panel of Fig. 2, for example, the direct baseline introduces a human face that is absent from our no-text background; re-prompting with a different subtitle would yield yet another face. This is not a failure of Qwen — it is a structural property of any pipeline that re-runs the generator per edit. Our pipeline never re-runs the generator after the background is accepted, so the scene is identical across edits up to typography rasterization. The SSIM and LPIPS values reflect this guarantee empirically; the qualitative pairs in Fig. 4 demonstrate it visually.

Table 4. Ablations. The contrast-aware renderer improves text readability, while saliency-aware layout reduces background clutter under text boxes.

Variant	OCR Exact	Word Recall	Contrast	Edge Density
Full	0.667	0.854	0.676	0.316
No saliency	0.700	0.843	0.677	0.328
No contrast	0.600	0.816	0.602	0.316

The second claim is that the poster-data layout prior is useful for practical typography placement. The GenPoster-trained layout model reaches 0.975 validation accuracy over real and negative text boxes, and the planner produced a valid in-bounds non-overlapping configuration for every prompt in the 24-prompt set, i.e. the search never failed to find a feasible layout under our spatial constraints. The direct T2I baseline in Figure 2 often creates plausible poster-like raster text, but it also adds unwanted labels, small glyphs, or “4K”-style artifacts and provides no editable layout.

The third claim is style diversity through tool routing. The router reaches 0.789 validation accuracy against weak GenPoster-derived labels, compared with a 0.398 majority-class baseline. Figure 3 shows the resulting visual variety: the system routes prompts to display fonts and effects such as condensed travel type, neon/retro shadows, editorial serif, Bauhaus geometry, luxury serif, and sport block lettering.

6. Ablations

We compare the full system to two earlier variants on the original ten-prompt diagnostic subset: removing the edge-density saliency term and disabling high-contrast color selection. Table 4 summarizes the diagnostic OCR results from the environment where Tesseract was available. The no-contrast variant reduces OCR exact match from 0.667 to 0.600 and lowers the contrast proxy from 0.676 to 0.602. The no-saliency variant places text over slightly higher-edge regions on average, increasing mean text-region edge density from 0.316 to 0.328. Although its OCR exact match is slightly higher on this small sample, its placements are often less compositionally balanced and more likely to interfere with salient content.

7. Supplementary Diagnostics

Figure 6 gives the supplementary confusion matrix requested for the tool router. The main confusions reflect the weak-label construction: modern minimal, cinematic, playful social, and retro pop are broad buckets, while rare classes such as space age and brutalist type have very lim-

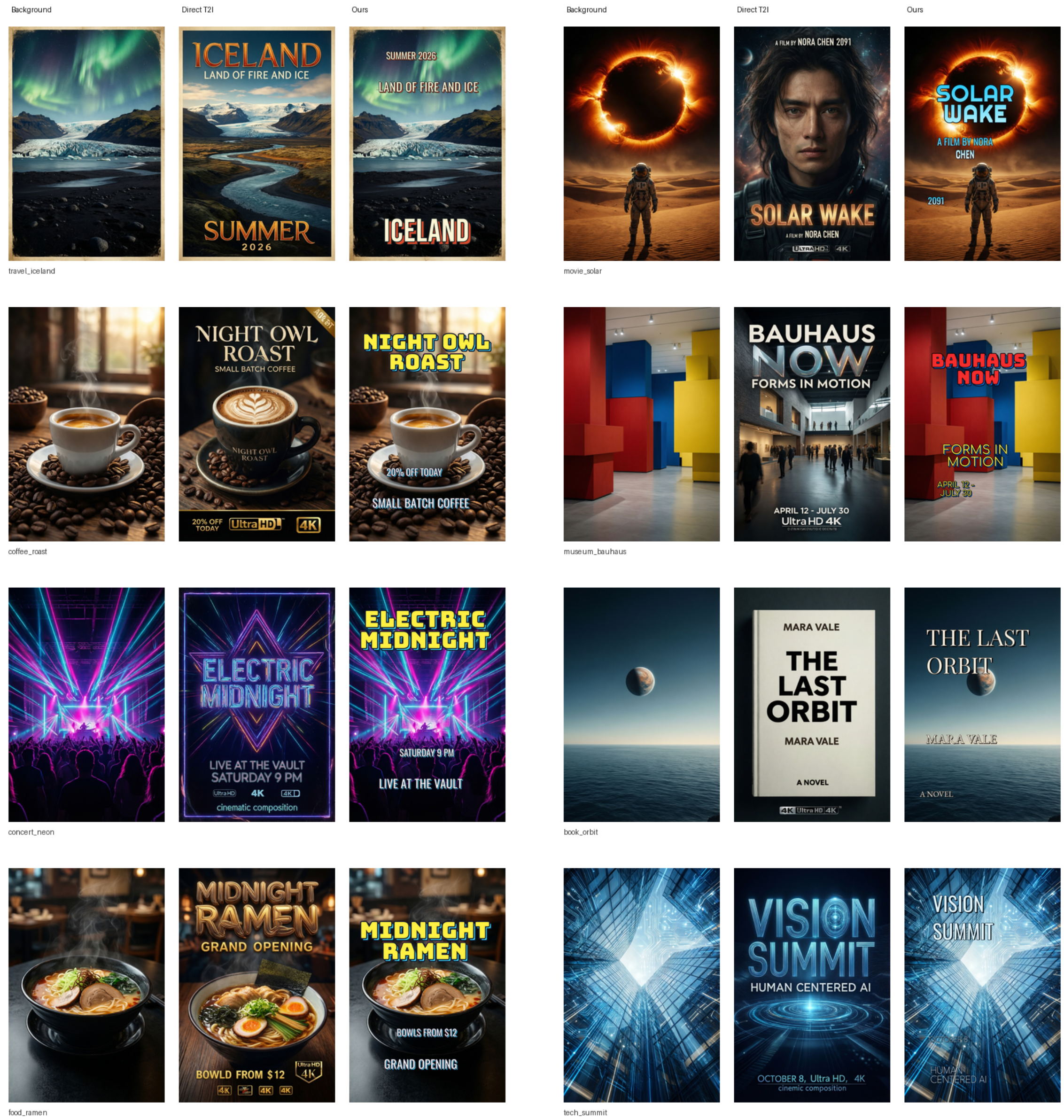

Figure 2. **Qualitative comparison.** Each group shows a no-text Qwen background, a direct Qwen T2I poster baseline, and our layered result. Direct T2I baselines are visually strong but often introduce extra text, pseudo-text, or non-editable raster typography. Our method uses generated backgrounds but renders the requested typography as a structured layer.

ited support. This supports our interpretation of the router as a diversity controller rather than a definitive style classifier.

8. Limitations

Our method is a layered design system rather than a fully end-to-end diffusion model. It does not learn kerning, curved text, custom font generation, or complex multi-line hierarchy at the level of a professional design tool. The Qwen background generator can occasionally insert unwanted text even when the background prompt forbids it, so strict no-text background generation remains imperfect. We note that this caveat applies only to the initial background; once the background is accepted, the no-regeneration property holds for all subsequent edits regardless of imperfections in the seed image. The GenPoster style labels are weakly supervised from metadata and prompt heuristics rather than human aesthetic preferences, so the router is best

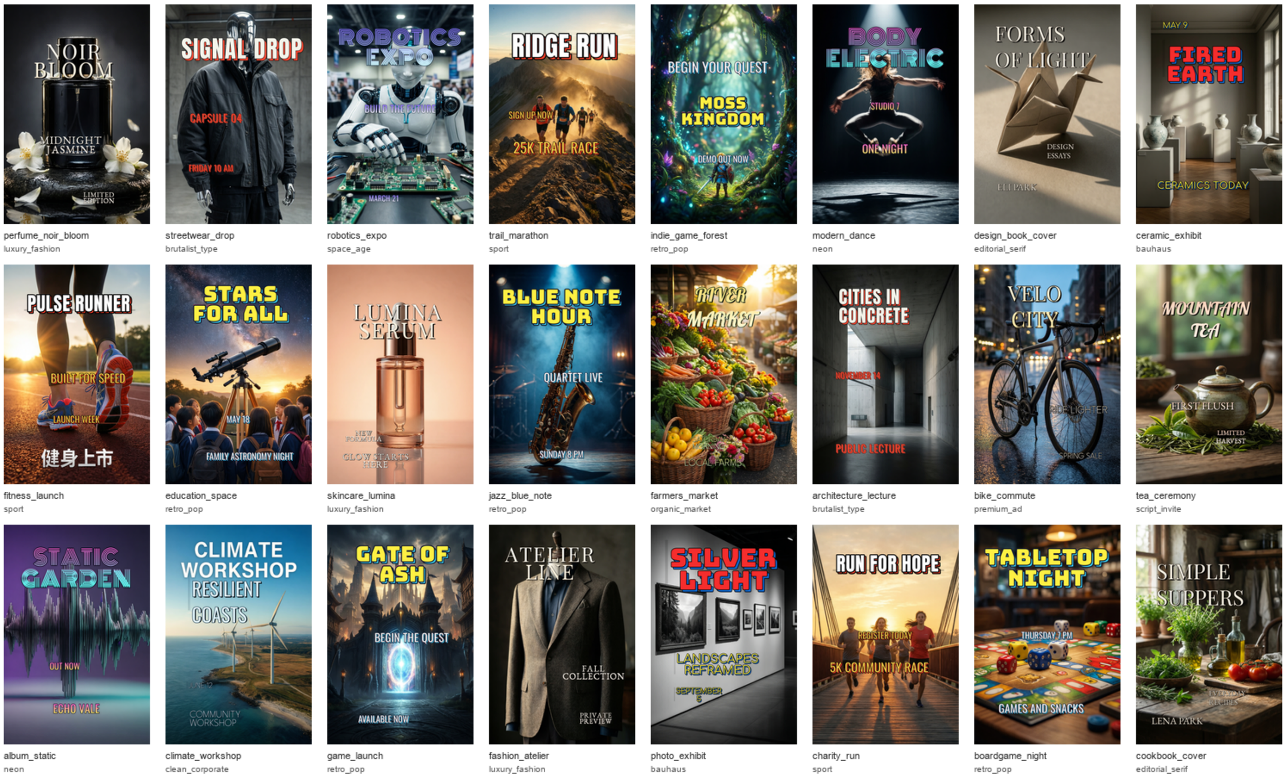

Figure 3. **Typography-tool routing.** This gallery uses prompts disjoint from Fig. 2. The learned router selects different font families, layout templates, and display effects for different poster categories, avoiding the single-style black-band appearance of the first prototype.

understood as a diversity controller rather than a definitive design critic. OCR is also an imperfect proxy: Tesseract can miss words that are visually correct to a human reader, especially when text is small, stylized, or placed on textured backgrounds. Still, for time-limited text-heavy design tasks, the layered approach offers a reliable tradeoff between visual quality, exact text, style control, and editability.

9. Conclusion

We presented a typography-aware poster generation pipeline that separates background synthesis from editable text rendering. The approach supports stable edits without background regeneration and avoids asking a T2I model to rasterize exact, editable typography. A lightweight trained layout ranker, a GenPoster layout prior, and a typography-tool router provide learned control over placement and visual style, while ablations show how saliency and contrast cues contribute to practical typography placement. More broadly, the project suggests that text-heavy visual synthesis should expose structured, editable layers rather than treating all visual content as a single raster generated from a prompt.

10. Use of Generative AI

Generative AI tools were used to help draft and revise prose, organize the CVPR-style report structure, and debug code/LaTeX. The experiments, scripts, generated figures, metrics, and final edits were run and verified in the project workspace.

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Figure 4. **Text editing.** Since text is a structured layer, changing a subtitle, date, or callout does not require regenerating the background.

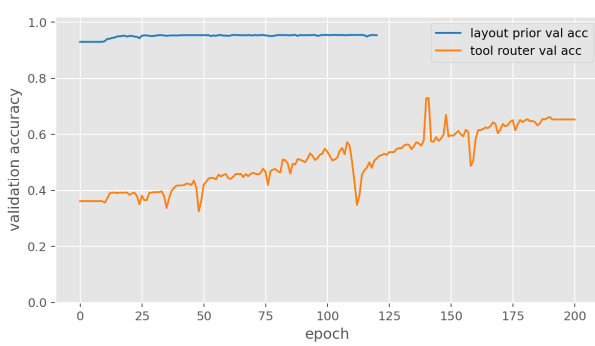

Figure 5. **Training curves** for the GenPoster layout prior and the typography-tool router. The layout prior converges quickly to high validation accuracy; the router accuracy is bounded by the noise of weak GenPoster-derived style labels.

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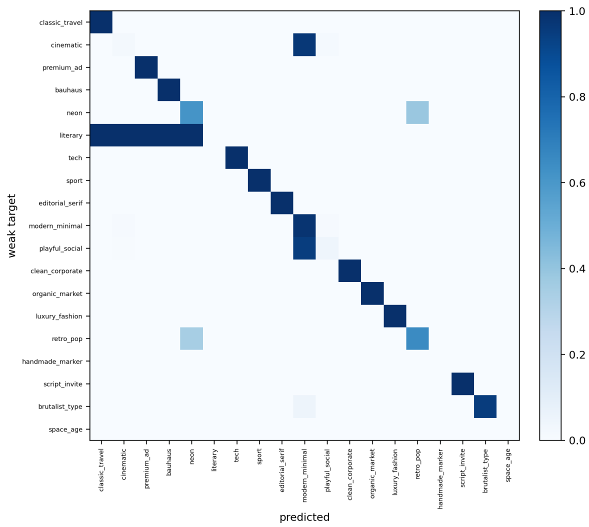

Figure 6. **Typography-tool router confusion matrix.** Rows are weak GenPoster-derived target styles and columns are predicted tool presets. The router is not a human aesthetic judge; it is a compact controller that maps prompt/design cues to reusable typography tools.

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